Acme Semiconductor, Inc.

XC-1000 Technical Reference Manual

XC-1000R / XC-1000E Preliminary Design Specification

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Author	Alice Chen, Bob Li
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Internal Use Only

Revision History

Version	Date	Author	Description of Change
v0.1	2026-08-01	A. Chen	Initial draft: system architecture and register map
v0.2	2026-09-05	B. Li	Added UART, clock, and reset sections
v0.3	2026-10-19	A. Chen	Refined UART parameter table and bit-field descriptions

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1 Overview

XC-1000 is a 32-bit RISC-V microcontroller for low-power IoT endpoints. It integrates a vector processing unit and runs at up to 400 MHz. This manual is intended for IC design, verification, and FPGA prototyping engineers, and describes the functions, registers, timing, and electrical characteristics of each subsystem.

1.1 Key Features

- 32-bit RISC-V RV32IMAC core at up to 400 MHz, with a 128-bit SIMD vector unit
- 256 KB SRAM and an 8-channel DMA controller on a multi-layer AHB/APB bus
- Three UARTs, two SPI interfaces, four timers, and a 32 kHz wake-up domain
- 1.8–3.6 V supply, QFN-48 package, and −40 °C to +105 °C operating range
- JTAG/cJTAG debug, watchdog, and brown-out reset

1.2 Terms and Abbreviations

Abbreviation	Description
AHB	Advanced High-performance Bus
APB	Advanced Peripheral Bus
cJTAG	Compact JTAG
DMA	Direct Memory Access
FIFO	First In, First Out
FPGA	Field-Programmable Gate Array
GPIO	General-Purpose Input/Output
IC	Integrated Circuit
IoT	Internet of Things
JTAG	Joint Test Action Group
PLL	Phase-Locked Loop
QFN	Quad Flat No-leads
RC	Resistor-Capacitor
RTL	Register Transfer Level
SIMD	Single Instruction Multiple Data
SoC	System on Chip
SPI	Serial Peripheral Interface
SRAM	Static Random Access Memory
UART	Universal Asynchronous Receiver/Transmitter

2 System Architecture

2.1 Block Diagram

See Figure 2-1. XC-1000 uses a multi-layer bus and multiple clock domains: the core domain runs on `hclk` (up to 400 MHz), the peripheral domain on `pclk`, and the low-power wake-up domain on a 32 kHz `lclk`.

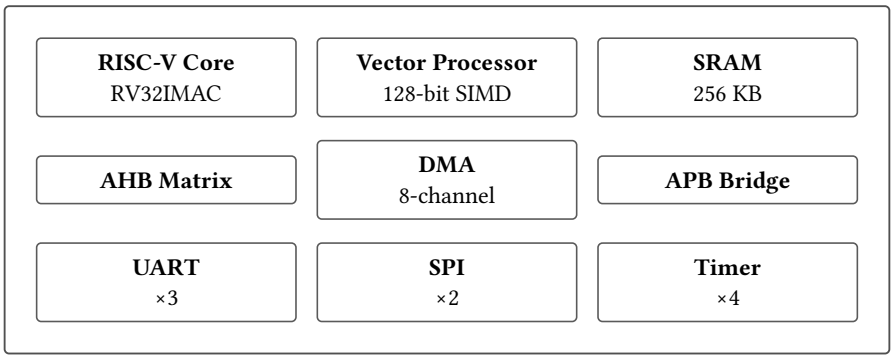


Figure 2-1: XC-1000 block diagram

2.2 Clock & Reset

The clock tree supports an external crystal (`hse`) and an internal RC oscillator. A PLL multiplies the reference clock for the core domain. Reset sources are listed in Table 2-1.

Table 2-1: Reset sources and behavior

Source	Scope	Behavior
`por_rstn`	Chip-wide	Power-on reset; asserted asynchronously and released synchronously after 16 `hclk` cycles
`wdt_rstn`	Core/peripheral	Triggered by a watchdog timeout; the wake-up domain is not reset
`sw_rstn`	Software	Selects subsystems to reset via `SYS_RSTCTL`

2.3 Design Constraints

Timing closure targets 400 MHz in the core domain (TSMC 28 nm, 0.9 V, 125 °C corner). The critical path lies in the multiply-accumulate array of the vector unit. Static timing margin must be at least 10%, and scan coverage at least 98%.

3 Registers

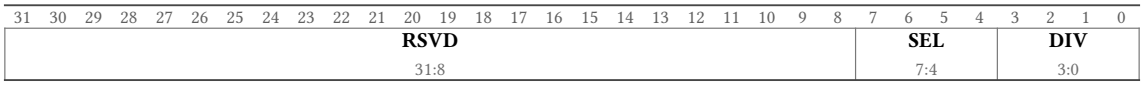
3.1 Register Map

The system controller base address is `0x4000_0000`. The register map is shown in Table 3-1.

Table 3-1: System controller register map (base address 0x4000_0000)

Register	Offset	Width	Access	Description
SYS_ID	0x000	32	RO	Chip ID and revision
SYS_RSTCTL	0x004	32	RW	Software reset control
SYS_CLKSEL	0x008	32	RW	Clock source and divider selection
SYS_INTEN	0x00C	32	RW	Interrupt enable
SYS_INTSTAT	0x010	32	RC_W1C	Interrupt status; write 1 to clear

3.2 Bit Fields



Field	Bits	Access	Reset	Description
RSVD	31:8	RO	0x0	Reserved; reads as 0, writes ignored
SEL	7:4	RW	0x1	System clock source: 0 = hse, 1 = pll, 2 = rc
DIV	3:0	RW	0x0	Clock divider: DIV + 1, giving 1–16

Figure 3-1: SYS_CLKSEL bit fields (offset 0x008)

4 Functional Description

4.1 UART Module

The UART supports programmable baud rates up to 6 Mbps, 8 or 9 data bits, hardware flow control, and a 64-byte FIFO. The baud rate is given by Equation (4-1).

$$\text{baud} = \frac{f_{\text{pclk}}}{16\,(N + \text{DL}/32)} \tag{4-1}$$

where N is the integer divisor and DL the fractional divisor; see Table 4-1.

Table 4-1: UART baud-rate parameters (`f_pclk` = 100 MHz)

Symbol	Parameter	Min	Typ	Max	Unit	Notes
f_{baud}	Baud rate	—	—	6	Mbps	Set by `pclk` and divider
Δf_{baud}	Baud-rate error	−2.0	—	+2.0	%	Over the operating temperature range
t_{bit}	Bit time	—	—	—	—	16× oversampling
N	Integer divisor	1	—	65,535	—	—

4.2 Firmware Example

The standard clock-switch sequence is as follows.

```
// Switch the system clock to PLL and wait for lock
sys_clksel_write(SYS_CLKSEL_SEL_PLL);
while (!sys_pll_locked()) {
    /* Wait for SEL to take effect; fall back to hse on timeout */
}
```

Appendix A: Package & Pinout

The device is supplied in a QFN-48 package measuring 7 mm × 7 mm, with a 0.4 mm pitch and an exposed 5 mm × 5 mm thermal pad. Table A-1 lists an excerpt of the pin assignments; the complete pin-multiplexing table is provided in the companion datasheet.

Table A-1: Pin assignments (excerpt)

Pin	Name	Type	Description
1	VDD	Power	1.8–3.6 V supply
2	VSS	Ground	Ground reference
3	POR_N	Input	Power-on reset, active low
4	HSE_IN	Input	Crystal oscillator input
5	HSE_OUT	Output	Crystal oscillator output
6	TCK	Input	JTAG test clock
7	TMS	Input	JTAG test mode select
8	UART0_TX	Output	UART0 transmit
9	UART0_RX	Input	UART0 receive
10	GPIO0	I/O	General-purpose I/O

References

[1] RISC-V International, “The RISC-V Instruction Set Manual, Volume I: Unprivileged ISA,” version 20240411, 2024.
[2] Arm Limited, “AMBA AHB-Lite Protocol Specification,” ARM IHI 0033.
[3] Arm Limited, “AMBA APB Protocol Specification,” ARM IHI 0024.
[4] Acme Semiconductor, “XC-1000 Datasheet,” 2026.